

Question				Marking Details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	i		60 000 : 1 Accept correct ratio not to 1 e.g. 120 000:2		1		1	1	
		ii		<i>Advantage:</i> Any one from: Reduces water loss/ allows them to live in arid conditions (1) No {blood / circulatory system/ pigment} required (1) Oxygen supplied directly to the cells/ muscles (1) Tracheoles go directly into cells/ tissues (1) <i>Disadvantage:</i> Size/ shape limitation (1)	2			2		
	(b)			Any 3 (x1) from: • Diffusion of gases related to Surface Area (1) • Oxygen use related to volume (1) • Amoeba has large Surface Area : volume ratio (1) • Therefore diffusion of gases sufficient (to supply demand) (1) • Short diffusion distance (1)	3			3		
	(c)			Ventilation (movements/system)/ description of replacing oxygen in alveoli/ ORA for carbon dioxide (1) {Blood/ transport system} (takes oxygen away from respiratory surface)/ ORA for carbon dioxide/ or description of (1)	2			2		